AARAV JAIN
2008aaravjain@gmail.com | +919999914123

EDUCATION

Step by Step School, Noida, India **2011-Present**
- International Baccalaureate Diploma program, Class 12; 2025-27
- Cambridge IGCSE board Examinations, Grade 10; 2023-25

Stanford ULO Pre-Collegiate University Level Math **2026**

STANDARDISED TESTS AND AP's

SAT Total- 1540 Mathematics-790; English- 750 **2026**
IELTS -Overall score: 8 **2026**
AP Examinations: Psychology- 5; Environmental Science -4 **2025**

ACADEMIC HONOURS AND AWARDS

School Awards:
- Award of Excellence in English Language & Literature and Mathematics AA HL 2025-26
- Certificate of Distinction in IGCSE 2023-25
- Award of Excellence in Physics, History, First Language English, and Spanish 2024-25
- Award of Excellence in Physics and English Literature 2023-24
- Consistent Academic Achievement Award 2022-23
- Outstanding Sportsperson of the year (Swimming) 2024-25

International Awards:
- Imperial College London STEM-a-thon: National and International winner- Plarick- A recycled plastic brick project 2025
- Gold CREST Award for Innovation and Sustainability: designed and built low-cost smart-irrigation system. 2025
- Blue Ocean Student Entrepreneur Competition: Top 250 (~1%) for Mindrive, an assistive mobility concept. 2025
- University of Waterloo mathematics Contest:Distinction and highest in school -Pascal, Fermat and Gauss 2026
- John Locke Essay Competition-Commendation Award- "Does Foreign Aid Help or Hurt Poor People" 2026

CLEAN WATER INITIATIVES

Arsenic, Lead, Heavy Metal and Turbidity Filtration

Innovation: **2025-26**
- Studied arsenic-remediation chemistry under Prof. Robin Datta at Tezpur University (Assam).Designed a low-cost portable bottle to filter arsenic other heavy metals and turbidity. Filed for patent -Application No. 202531130896
- Developed an automated community arsenic filtration system using tank-level sensors, IoT communication.

Outreach: **2025**
- Partnered with Govt. of Assam to distribute 250+low-cost portable bottle filter for arsenic, providing safer drinking water to 1,250+ residents. Set up an auto chemical dosing community system serving approximately 2500 villagers/daily.

FarmTech for Rural India

Smart Irrigation Device **2025-26**

Innovation:
- Designed an Arduino and ESP 32-based irrigation system using soil-moisture sensors that dispenses water intelligently

Outreach: **2025 and 2026**

Deployed the Smart Irrigation System across 35 farms in Assam and Haryana
- Partnered with the Govt. of Assam to deploy across 20 farms in Assam; reported a 20% reduction in water consumption.
- Partnered with the Indigo Knowledge Prism Foundation to establish a village research-lab initiative, training 15 children from farming families to build, test, and deploy the system in their own farmlands. Installed in 15 farmlands in Haryana

Commercial Pursuit: **2026**
- Developed a business plan; sold the construction model to a private company for INR 70,000.

Microplastics

Research: **2026**

Author, Bioinformatics-Guided Comparative Analysis of Moringa and Tamarind for Microplastic Removal
Manuscript accepted to be published in International Journal of High School Research
- Compared Moringa coagulant protein and tamarind xyloglucan against PP, PE, PS and PET using AutoDock Vina and SMINA; strongest binding was Moringa–PET (−5.015 kcal/mol).

Outreach: **2026**

Microplastics free water
- In collaboration with Kuda Bazar Foundation established a community auto dosing unit using Moringa Coagulant Protein to filter microplastics form water. Impacted 1000s of people per day.

Lead Filtration

Research: **2026**

Author, Comparative Evaluation of Lead Removal from Contaminated Water by Chemical Precipitation Using Different Reagents under Varying pH and Dosage Conditions.

Manuscript published in Oxford Journal of Student Scholarship

- Conducted study testing 27 combinations comparing sodium carbonate, sodium sulphate, and sodium hydroxide for lead removal with varying pH and dosage conditions. Found sodium carbonate the strongest precipitant (92.77% removal)

PLASTIC RECYCLING 2025-26

PLARICK- Built low-cost, hollow interlocking bricks from recycled plastic for temporary and emergency housing; developed a business plan and received a funding offer. Earned the Imperial College London Stem-a-thon-National and International winner award.

STEM PURSUITS

Internships and field studies:

Renewable-Energy Study, Waaree Energies Ltd. -Chikli (Maharashtra) **2026**

- Learned wafer-to-PERC/TOPCon cell production, automated module assembly. Produced a report connecting material science, photovoltaic engineering and manufacturing.

Effluent Waste Water Treatment Plant **2026**

- Studied wastewater treatment plant; applied findings to microplastic and lead removal research.

CPVC Plumbing Pipe & Fittings Plant, Pune **2025**

- Gained exposure to manufacturing processes and studied how pipe geometries are engineered for different applications.

Mentoring:

Teaching Assistant, Young Technology Scholars (YTS), Plaksha University **2026**

- Selected as the programme's only high-school TA after previously attending as a Scholar in 2024; mentored 48 students across 12 teams designing and building weather-sensing gliders under Prof. Rajkumar S. Pant from IIT Bombay.

Founder, Science That Wins Matches **2025-2026**

- Authored the book - "Science That Wins Matches" designed STEM curriculum; reached 10,000+ students across 3 states
- Led hands on science sessions for 150 children; trained 16 student mentors enabling more learners through peer mentoring

FarmTech for Rural India – Smart Irrigation Device **2025-2026**

- Partnered with the Indigo Knowledge Prism Foundation to establish a village research-lab initiative, training children from farming families to build, test, and maintain devices independently, witnessing 30% reduction in water consumption.
- Trained 15 students from farming families in Jalota (Haryana) to build smart irrigation devices.

***Enrichment Programmes**:*

Imperial College London — Global Summer Programme **2025**

- Learned engineering design, prototyping, and iterative problem-solving, applied learning to independent projects

Young Technology Scholars (YTS), Plaksha University — Summer Scholar **2024**

- Completed YTS's two week programme, building a soil moisture sensor; independently developed it over six months

Fundamental Skills in Engineering Design, University of Leeds (Coursera) **2025**

- Mastered engineering design, prototyping and iterative testing; applied these to irrigation and filtration projects.

SCHOOL LEADERSHIP

Student Council - House vice-captain **2025**

- Lead house participation across academic, sporting and cultural inter-school tournaments

Founder and Head Debate Society **2022- 2026**

- Founded the school's first student-led debate society, built a 7 member executive team
- Designed selection for 350+applicants, mentored 30 members through weekly workshops and training
- Organised "Elenchus", the first student-run interschool debate tournament with 16 participating schools;100+participants

URJA School STEM Festival- Core Committee Member **2024 - 2026**

- Launched three engineering events – Robo Race, Paper Plane Challenge and Science Exhibition engaging 300+students

Core Production Team, Jashn — Annual Performing & Visual Arts Festival **2026**

- Served on the Core Production Team, leading event logistics for a school-wide arts festival with 600+ participants.

DEBATE ACHIEVEMENTS 2022-2026

Represented the school and won multiple awards at 20+ national and international debate tournaments like ISDS National Camp (top 20), winner at SRCC Nationals, ISDS Iron Person Triathlon top 20, Doxbridge quarterfinalist, Yale Invitational -top 10 speaker, Asia School Open – top 10 speaker and others.

EXTRACURRICULAR ACTIVITIES

First Tech Challenge **2020-24**

- FTC: won national Runner-Up Innovate Award, FLL: qualified for National Championship twice

Competitive Swimming **2020-26**

- Competed on the school swim team across 10+ interschool, state, and national-level meets over seven years
- Won 2 Gold and 1 Silver, ISSO Swimming Championship, ranked 12th in the 100m Breaststroke at the Asia Open Schools Invitational, Bangkok, Awarded Outstanding Sportsperson of the Year — Swimming by the school

SKILLS AND INTERESTS

- Proficient in English Spanish and Hindi
- Laboratory and research tools: Atomic Absorption Spectroscopy, AutoDock Vina, SMINA, PyMOL, SwissADME
- Comfortable utilizing research tools such as Google Scholar and citation management software